

Employment

Product Engineer Intern	Isomer AI	May 2025 – May 2026
- Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 923 commits across 7 production deployments. - Engineered agentic AI workflows (Claude, GPT-4o, Gemini) via MCP with provider-agnostic interfaces; built document processing (PDF gen, email parsing, S3) and full Playwright test coverage.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deployed cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling interactive exploration of 500GB+ strawberry and grape datasets for 100+ researchers worldwide. - Improved ML model accuracy by 12% by applying hyperparameter tuning and time-series feature engineering on biological sensor data collected from genomic field experiments.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated cloud deployments across AWS and GCP via CI/CD pipelines; integrated Grafana monitoring dashboards and led model validation and red-teaming sessions to improve system reliability across 5+ production services.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led end-to-end SDLC for a data-driven system automating support ticket analysis using NLP classification models; reduced average resolution time by 15% and eliminated manual triage for 2,000+ monthly tickets. - Implemented distributed data pipelines using Kafka streaming and Spark + Hadoop batch processing for large-scale ticket ingestion; reduced end-to-end ticket-processing latency by 20% at peak load.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

Glance AI	Privacy-First RAG	2025
- Designed per-tenant RAG pipeline for healthcare with strict vector isolation preventing cross-tenant data leakage and LLM hallucinations. - Implemented deterministic retrieval for regulated environments; eliminated context bleed between tenants for factual LLM outputs.		
AlgoPulse	Agentic AI Platform	2025
Replaced API-polling bottleneck with Supabase Edge Functions streaming AI reasoning loops to React Native clients in real time; Temporal.io durable workflows; 2,000 async inference req/sec under sustained load.		
PerCent	GenAI Finance App – HackNC24 Top 10	2024
- Built GenAI finance app with goal tracking, expense logging, and AI chatbot in React Native; ranked Top 10 of 100+ teams in 48-hour sprint. - Designed RAG-based chatbot using Selenium for real-time financial data retrieval, Hugging Face embeddings, and Qwen-14B-Chat inference; reduced manual financial research time by 30% per user session.		

Languages and Technologies

- Languages:** Python, TypeScript, JavaScript, Java, SQL, Bash
- Frontend & Full Stack:** React, React Native, Node.js, Express, REST APIs, PostgreSQL, Redis, Auth0
- AI/ML & Healthcare:** Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, Hugging Face, OCR/NLP pipelines
- Cloud & DevOps:** AWS, GCP, Docker, Temporal.io, Terraform, GitHub Actions, Grafana, Sentry

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog